

Question			Marking Details	Marks Available					
				AO1	AO2	AO3	TOTAL	Maths	Prac
1	(a)	i	8 (ms) = 2 marks Award 1 mark for: 0.0083 1/120 8.3 8.33		2		2	2	
		ii	<u>Threshold</u> was not reached (so no action potential) (1) Not enough {sodium ions/ Na ⁺ } entered the {neurone/ axon}(1) Reject sodium unqualified		2		2		
	(b)	i	Schwann (cell)	1			1		
		ii	Lipid/ fat	1			1		
		iii	5.15×10^{-5} s = 3 marks 0.0000515 s = 2 marks $5\ 150/1 \times 10^8$ = 1 mark		3		3	3	
	(c)	i	Any four (×1) from A. Rate would increase (1) B. Because more <u>aerobic</u> respiration (1) C. Saltatory conduction would not occur/ Whole membrane would need to be depolarised/ action potentials would occur at more places along neurone/ shorter local circuits occur (1) D. so more {Na ⁺ /K ⁺ } pumps (have to work across whole length of exposed membrane)/ more active transport(1) E. More ATP needed (to maintain resting potential) (1)			4	4		
		ii	Any two (×1) from: Person would feel tired/ exhaustion/ feel weaker (1) Lack of feeling/ OWTTE (1) {Longer/ slower} reaction times (1) paralysis/ no response/ loss of muscle control (1)			2	2		
			QUESTION 1 TOTAL	2	7	6	15	5	0